MPG No. 10.3.34 $\bar{4}'$ [Type III, tetragonal] [M tensor]

* Rank 0 tensor.

$$[M]$$

$$M = \mathbb{M}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & M_{xx} & 0 \\ 0 & 0 & M_{zz} \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)}$$
$$M_{zz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & -M_{xxz} & 0 \\ M_{xxz} & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{\beta(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & \frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & M_{yzx} & 0 \\ 0 & 0 & -2M_{yzx} \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yzx} = \mathbb{T}_v^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{T}_{xy}^{(1)}$$
$$2M_{yzx} = 2\mathbb{T}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & 0 \\ M_{yzy} & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_{v}^{(2)}$$
$$M_{yzy} = \mathbb{T}_{xy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & 0 & 0 & -M_{xxxy} \\ M_{xxzz} & M_{xxzz} & M_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ M_{xxxy} & -M_{xxxy} & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} - \mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_{4}^{(1)} - 2\mathbb{M}_{u}^{(1)} - 4\mathbb{M}_{u}^{(2)}$$
$$M_{xxyy} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4}^{(1)} - 2\mathbb{M}_{u}^{(1)}$$
$$M_{xxzz} = \mathbb{M}_{0}^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4}^{(1)} + \mathbb{M}_{u}^{(1)}$$
$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)}$$
$$M_{zzzz} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_{4}^{(1)} + 4\mathbb{M}_{u}^{(1)} + 8\mathbb{M}_{u}^{(2)}$$
$$M_{yzyz} = \mathbb{M}_{0}^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4}^{(1)} + \mathbb{M}_{u}^{(2)}$$
$$M_{xyxy} = \mathbb{M}_{0}^{(2)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4}^{(1)} - 2\mathbb{M}_{u}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & M_{xxzz} & 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{xxzz} & 0 & 0 & -M_{xxxy} \\ -M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & M_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxzz} = 3\mathbb{M}_{u}^{(3)}$$
$$M_{xxxy} = 2\mathbb{T}_{z}^{(1)} - 2\mathbb{T}_{z}^{\alpha(1)}$$
$$M_{yzzx} = -\mathbb{T}_{z}^{(1)} - 4\mathbb{T}_{z}^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)}$$
$$M_{xyxy} = \mathbb{M}_{0}^{(3)} + 4\mathbb{M}_{u}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_{z}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ M_{yzyz} & M_{yzzx} & 0 \\ M_{yzzx} & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)}$$
$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$
$$M_{yzyz} = -3\mathbb{M}_u^{(5)}$$
$$M_{yzzx} = \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = -3\mathbb{M}_u^{(6)}$$
$$M_{yzzx} = \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & 0 & 0 & -M_{xxyx} & 0 & 0 & -M_{xxxy} \\ M_{zzxx} & M_{zzxx} & M_{zzzz} & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzy} & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & M_{yzzy} & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{xxzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)}$$
$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)}$$
$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}$$
$$M_{zzxx} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)}$$
$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$
$$M_{yzyz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)}$$
$$M_{yzzx} = -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)}$$
$$M_{yzzy} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)}$$
$$M_{yzxz} = -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}$$
$$M_{xyxx} = \mathbb{M}_{4z}^{\alpha(1)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$
$$M_{xyxy} = \mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzy} & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & -M_{yzzy} & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)}$$
$$M_{yzzx} = -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{yzzy} = -\mathbb{M}_0^{(3)} + 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)}$$
$$M_{yzxz} = \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$
$$M_{xyxy} = \mathbb{M}_0^{(3)} + 4\mathbb{M}_u^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & M_{xxxy} & 0 \\ -M_{xxxy} & M_{xxxx} & 0 \\ 0 & 0 & M_{zzzz} \\ 0 & 0 & M_{yyzz} \\ M_{zzxx} & 0 & 0 \\ M_{xxxy} & M_{xxyy} & 0 \\ 0 & M_{yzzy} & 0 \\ 0 & 0 & M_{zxxz} \\ M_{xxyy} + M_{yyzz} - M_{yzzy} - M_{zxxz} + M_{zzxx} & -M_{xxxy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = 3\mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$
$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)}$$
$$M_{zzzz} = 3\mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{yyzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)}$$
$$M_{zzxx} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{yzzy} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)}$$
$$M_{zxxz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)}$$